

Researchers studied the effects of heart disease on basic indicators of health and fitness. A small sample of heart disease patients and healthy subjects underwent physiological tests to assess cardiovascular function. The results for individual subjects are shown in Tables 1 and 2.

Table 1 Test Results from Heart Disease Patients

Test	Subject ID		
	1a	1b	1c
Resting Heart Rate (bpm)	70	75	80
Distance walked in 20 s (m)	20	25	15
Maximum velocity during walk test (m/s)	3	2.5	1.5
Holding weight with outstretched arm (s)	10	9	7

Table 2 Test Results from Healthy Subjects

Test	Subject ID		
	2a	2b	2c
Resting Heart Rate (bpm)	65	72	70
Distance walked in 20 s (m)	30	25	29
Maximum velocity during walk test (m/s)	2.2	2.5	2
Holding weight with outstretched arm (s)	16	12	13

During the walk test, the subject's position along the track and their velocity were monitored with a laser Doppler sensor. For the weight test, subjects lifted a 3 kg object 20 cm off a table and held it in front of them at arm's length for as long as they could. The speed of light is 3×10^8 m/s, and gravitational acceleration is 10 m/s^2 .

The researchers want to measure the friction properties of the walking track using a block and a force gauge. They apply a force to the block, causing it to slide at a constant velocity along the track. What information is required to calculate the coefficient of kinetic friction for the track surface?

- ☐ A. Mass of the block, magnitude of the force, contact area between block and track [18%]
- ☐ B. Mass of the block, magnitude of the force, density of the block [1%]
- ☒ C. Mass of the block, magnitude of the force [73%]
- ☐ D. Mass of the block, velocity of the block, contact area between block and track [5%]

Correct

74% answered correctly

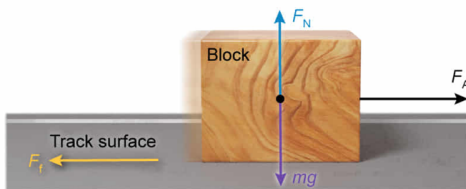
Explanation:

Determining the coefficient of kinetic friction for a track

$$\begin{aligned} F_f &= \mu_k F_N = \mu_k mg \\ F_A &= F_f \\ F_A &= \mu_k mg \end{aligned}$$

- F_f Friction force
- μ_k Coefficient of kinetic friction
- F_N Normal force
- m Mass
- g Gravitational acceleration
- F_A Applied force

Block does not move vertically, $\therefore F_N = mg$
Block slides horizontally with constant velocity, $\therefore F_A = F_f$
 F_A measured with force gauge



Calculating μ_k requires knowing F_A and m

$$\mu_k = \frac{F_A}{mg}$$

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Kinetic **friction** is the force that opposes the motion of an object as it slides on a surface. The kinetic friction force F_f is the product of the **coefficient of kinetic friction** μ_k and the **normal force** F_N exerted on the object by the surface. When an object moves horizontally along a flat surface, F_N is equal to the object's **weight**, which is the product of the mass m of the object and the gravitational acceleration g :

$$F_N = mg$$

Hence, F_f can be calculated as:

$$F_f = \mu_k F_N = \mu_k mg$$

In this question, **Newton's first law of motion** implies that the net force on the block is zero because the block moves at a constant velocity along the track surface.

Consequently, the force F_A applied to the block equals F_f :

$$F_A = F_f$$

$$F_A = \mu_k mg$$

Rearranging the above equation shows that μ_k is equal to F_A divided by the product of m and g :

$$\mu_k = \frac{F_A}{mg}$$

Therefore, the only information the researchers need to calculate μ_k is the mass m of the block and the magnitude of the force F_A applied.

(Choices A, B, and D) The value of μ_k does not depend on the contact area between the block and the surface or on the density or velocity of the block. Only the values of m and F_A are needed to calculate μ_k .

Educational objective:

An object sliding on a surface experiences a kinetic friction force that opposes the motion. If the object is sliding with constant velocity on a flat surface, the coefficient of kinetic friction is calculated from the force applied to the object and the mass of the object.

Time spent:0 sec

Subject:
Physics

QID:403230

System:
Mechanics and Energy

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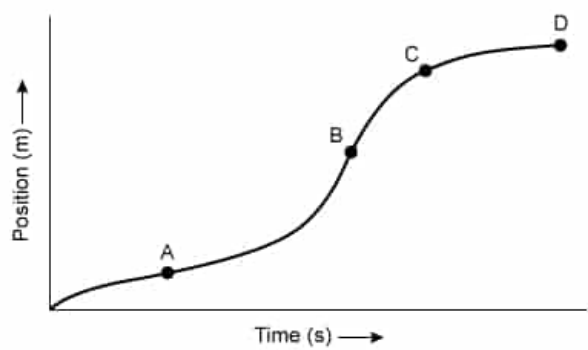
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The following graph plots the position of subject 2b during the walk test.



Which point on the graph corresponds to the greatest velocity of the subject during the test?

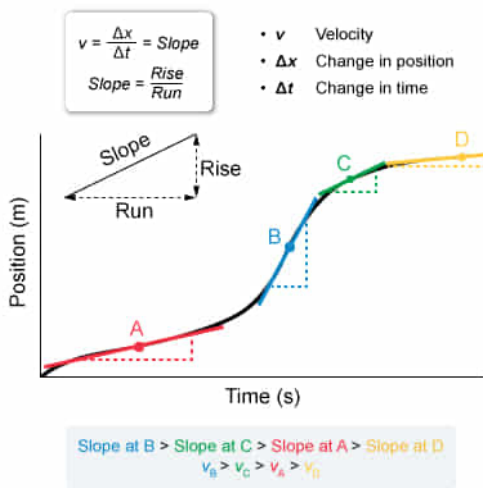
- ☐ A. Point A [2%]
- ✓ ☒ B. Point B [86%]
- ☐ C. Point C [7%]
- ☐ D. Point D [4%]

Correct

86% answered correctly

Explanation:

Determining velocity from plot of position vs. time



The **velocity** v of an object is defined as the change in the object's **position** Δx divided by the change in time Δt :

$$v = \frac{\Delta x}{\Delta t}$$

In a plot of position vs. time, the average velocity over a finite period is calculated by dividing Δx by Δt . However, the object's **instantaneous velocity** at an individual point in time is equal to the **slope** of the **tangent line** on the graph, which is the ratio of the line's vertical rise and horizontal run:

$$v = \text{slope} = \frac{\text{rise}}{\text{run}}$$

In this question, the plot of the subject's position vs. time has a small slope at the start near point A, indicating they had a low velocity at this time. The slope of the line increases at point B, reflecting an increased velocity. At the end of the graph around points C and D, the slope of the line decreases, showing that their velocity was reduced. Therefore, the subject's velocity was greatest at point B.

(Choices A, C, and D) The slopes of the line on the graph at points A, C, and D are less than at point B. Therefore, the subject's velocity was less at these points than it

was at point B.

Educational objective:

The slope of a plot of position vs. time is equal to the velocity. The velocity of the subject is greatest where the slope of the plot is greatest.

Time spent:0 sec

Subject:

Physics

QID:403231

System:

Mechanics and Energy

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Table 2 Test Results from Healthy Subjects

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The researchers want to increase the work done by the subjects during the weight test. Which of the following changes to the test will increase the work by 50%? (Note: Assume the object is lifted straight up off the table.)

- ☐ A. Increasing the mass of the object by 50% and the distance the object is lifted by 50% [11%]
- ☐ B. Increasing the mass of the object by 25% and the distance the object is lifted by 25% [43%]
- ☒ C. Increasing the mass of the object by 25% and the distance the object is lifted by 20% [33%]
- ☐ D. Increasing the mass of the object by 100% [11%]

Correct

34% answered correctly

Explanation:

Work required to lift an object

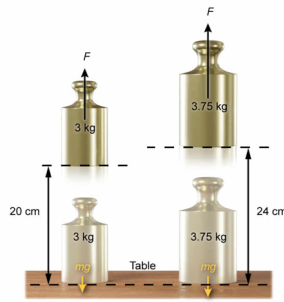
$$F = mg$$

$$W = Fd = mgd$$

- F Force
- m Mass
- g Gravitational acceleration
- W Work
- d Distance

$$W = (3 \text{ kg}) \left(10 \frac{\text{m}}{\text{s}^2} \right) (0.20 \text{ m}) = 6 \text{ J}$$

$$W = (3.75 \text{ kg}) \left(10 \frac{\text{m}}{\text{s}^2} \right) (0.24 \text{ m}) = 9 \text{ J}$$



Increase m from 3 kg to 3.75 kg (25% increase) & d from 20 cm to 24 cm (20% increase)
 W increases from 6 J to 9 J (50% increase)

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Lifting an object requires applying a **force** F equal to the object's **weight**, which is equal to the product of the object's mass m and the gravitational acceleration g :

$$F = mg$$

The **work** W done to lift the object is the product of F and the distance d the object is lifted:

$$W = Fd = mgd$$

In this question, the subjects lift an object with $m = 3 \text{ kg}$ over a distance $d = 20 \text{ cm}$. The work done by each subject is:

$$W = mgd = (3 \text{ kg}) \left(10 \frac{\text{m}}{\text{s}^2} \right) (20 \text{ cm})$$

Converting the units from centimeters to meters yields:

$$W = (3 \text{ kg}) \left(10 \frac{\text{m}}{\text{s}^2} \right) (20 \text{ cm}) \left(0.01 \frac{\text{m}}{\text{cm}} \right) = 6 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2} = 6 \text{ J}$$

The work should be increased by 50%, meaning the original value of 6 J should be multiplied by a factor of 1.50, yielding 9 J. Multiplying 3 kg by 1.25 (ie, 25% increase) and multiplying 20 cm by 1.20 (ie, 20% increase) gives:

$$W = (3 \text{ kg})(1.25) \left(10 \frac{\text{m}}{\text{s}^2} \right) (20 \text{ cm})(1.20) \left(0.01 \frac{\text{m}}{\text{cm}} \right) = 9 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2} = 9 \text{ J}$$

Therefore, increasing m by 25% and d by 20% results in a 50% increase in W .

(Choice A) Increasing m by 50% and d by 50% causes the work to increase by 125% to 13.5 J. This is greater than the 50% increase specified by the question.

(Choice B) Increasing m by 25% and d by 25% causes the work to increase by 56% to 9.4 J. This is greater than the 50% increase specified by the question.

(Choice D) Increasing m by 100% causes the work to increase by 100% to 12 J. This is greater than the 50% increase specified by the question.

Educational objective:

The work done to lift an object is the product of the object's mass, the gravitational acceleration, and the distance lifted. Changes to any of these parameters cause a proportionate change in the work done.

Time spent:0 sec

Subject:

Physics

QID:403232

System:

Mechanics and Energy

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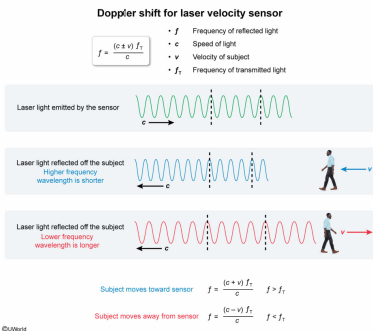
When a subject moves toward the laser Doppler sensor during the walk test, the laser light reflected back to the sensor will be:

- ☐ A. lower in velocity than the light transmitted by the sensor. [8%]
- ☐ B. higher in velocity than the light transmitted by the sensor. [9%]
- ☐ C. lower in frequency than the light transmitted by the sensor. [17%]
- ☒ D. higher in frequency than the light transmitted by the sensor. [63%]

Correct

63% answered correctly

Explanation:



Lasers emit **light** with a narrow range of **frequencies**. A laser **Doppler** velocity sensor detects the frequency of the light that reflects off an object. If the speed of light is c and the object moves with a velocity v , the reflected light will have a different frequency f than the light transmitted by the sensor f_T :

$$f = \frac{(c \pm v)f_T}{c}$$

When the object moves toward the sensor, the sum of c and v is used in the equation:

$$f = \frac{(c + v)f_T}{c}$$

In this situation, the light is shifted to a higher frequency:

$$f > f_T$$

When the object moves away from the sensor, the difference between c and v is used:

$$f = \frac{(c - v)f_T}{c}$$

The light is shifted to a lower frequency for this case:

$$f < f_T$$

In this question, the subject moves toward the sensor during the walking test. Therefore, the light reflected off the subject will have a higher frequency than the light that was transmitted by the sensor.

(Choices A and B) The velocity of light is a constant and will not change due to the subject's movements.

(Choice C) The light reflected back will have a lower frequency than the transmitted light when the subject moves away from the sensor. However, the question specifies the subject moves toward the sensor.

Educational objective:

The Doppler effect can be used to measure the velocity of a test subject. If the subject moves toward the sensor, the light reflected back to the sensor will have a higher frequency than the light originally transmitted.

Time spent: 0 sec
Subject:
 Physics

QID: 403233
System:
 Mechanics and Energy

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During the weight test, a subject lifts the object 15 cm off the table, holds it motionless for a moment, and then drops it. What is the velocity of the object immediately before it lands on the table? (Note: Assume air resistance is negligible.)

- ☐ A. 0.87 m/s [8%]
- ☒ B. 1.7 m/s [68%]
- ☐ C. 2.0 m/s [8%]
- ☐ D. 3.0 m/s [14%]

Correct
68% answered correctly

Explanation:

Velocity of a falling object dropped from rest:
Conversion of potential energy into kinetic energy

$PE_{\text{initial}} = mgh$
 $KE_{\text{final}} = \frac{1}{2}mv^2$
 $v = \sqrt{\frac{2KE_{\text{final}}}{m}}$

- PE_{initial} Initial potential energy
- m Mass
- g Gravitational acceleration
- h Height
- KE_{final} Final kinetic energy
- v Velocity

15 cm

3 kg

Table

Initial

$PE_{\text{initial}} = (3 \text{ kg}) \left(10 \frac{\text{m}}{\text{s}^2}\right) (0.15 \text{ m}) = 4.5 \text{ J}$

↓

Conservation of energy

$PE_{\text{initial}} = KE_{\text{final}} = 4.5 \text{ J}$

↓

Final

$KE_{\text{final}} = 4.5 \text{ J}$

$v = \sqrt{\frac{2(4.5 \text{ J})}{3 \text{ kg}}} = 1.7 \frac{\text{m}}{\text{s}}$

The law of conservation of energy states that the total mechanical energy E , which is the sum of potential energy PE and kinetic energy KE , must remain constant.

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$$E = PE + KE = \text{constant}$$

For example, the E of a falling object remains constant, but it is transformed from PE into KE as it falls:

$$E = PE_{\text{initial}} + KE_{\text{initial}} = PE_{\text{final}} + KE_{\text{final}}$$

The gravitational PE_g of the object-Earth system is the product of the object's mass m , the gravitational acceleration g , and the object's height h :

$$PE_g = mgh$$

Furthermore, the KE of an object is half the product of m and the object's velocity v squared:

$$KE = \frac{1}{2}mv^2$$

The value of v can be calculated from KE and m by rearranging terms:

$$v = \sqrt{\frac{2KE}{m}}$$

In this question, the 3 kg object at rest is dropped from a height of 15 cm. Consequently, KE_{initial} is zero and E is equal to PE_{initial} :

$$E = PE_{\text{initial}} = (3 \text{ kg})\left(10 \frac{\text{m}}{\text{s}^2}\right)(0.15 \text{ m}) = 4.5 \text{ J}$$

Immediately before the object lands on the table, h is zero, so PE_{final} is also zero; therefore, E equals KE_{final} . Because the PE decreased from 4.5 J to 0 J, KE_{final} must equal 4.5 J

$$E = KE_{\text{final}} = 4.5 \text{ J}$$

Substituting these values into the above equation for v yields:

$$v = \sqrt{\frac{2(4.5 \text{ J})}{3 \text{ kg}}} = \sqrt{3} \approx 1.7 \frac{\text{m}}{\text{s}}$$

(Choice A) If the $\frac{1}{2}$ factor is not included in the equation for KE , the value of v will be calculated incorrectly as 0.87 m/s.

(Choice C) If the value of h is incorrectly set to 20 cm instead of 15 cm, the value of v will be calculated incorrectly as 2.0 m/s.

(Choice D) If the square root is not included in the equation for v , the value of v will be calculated incorrectly as 3.0 m/s.

Educational objective:

The energy of a falling object is converted from potential energy into kinetic energy. The final velocity of the object is calculated based on its initial potential energy when it was dropped.

Time spent: 0 sec

Subject:
Physics

QID: 403234

System:
Mechanics and Energy

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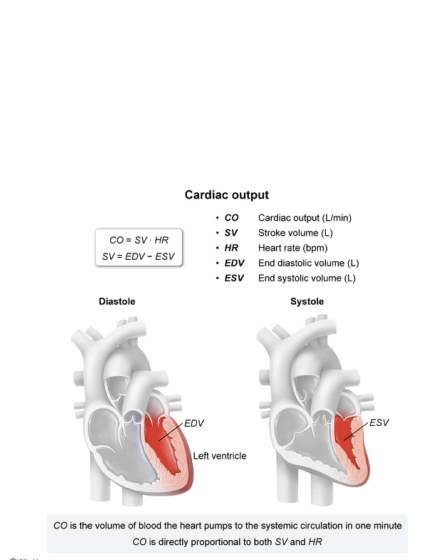
The cardiac output of a subject is directly proportional to their:

- ✓ ☒ A. heart rate. [78%]
☐ B. vascular resistance. [6%]
☐ C. left ventricle volume. [9%]
☐ D. total blood volume. [4%]

Correct

78% answered correctly

Explanation:



The **cardiac output** CO is the amount of blood pumped out of the left ventricle in 1 minute. CO is defined as the product of the stroke volume SV of the heart and the **heart rate** HR:

$$CO = SV \cdot HR$$

The SV is the volume of blood pumped out in a single heartbeat. The SV depends on the change in the volume of the left ventricle before and after the contraction of the heart. The volume of the left ventricle before contraction is the end diastolic volume EDV and the volume after contraction is the end systolic volume ESV , yielding:

$$SV = EDV - ESV$$

In this question, HR is the only answer choice that is included in the equation for CO . Based on this equation, CO is **directly proportional** to HR because any changes in HR will lead to an equivalent change in CO .

(Choice B) CO can be calculated based on the vascular resistance. However, CO and resistance are inversely proportional, not directly proportional.

(Choice C) The volume of the left ventricle is not a constant value and can influence CO in different ways. Increasing the EDV could increase the SV and CO , but a greater ESV will decrease the SV and CO .

(Choice D) The total blood volume does not influence CO .

Educational objective:

The cardiac output is the volume of blood the heart pumps in 1 minute. It is calculated by multiplying the stroke volume of the heart and the heart rate.

Time spent: 0 sec

Subject:
Physics

QID: 403235

System:
Mechanics and Energy